

Cybersecurity Terms with Definitions and Examples

Term	Definition	Example
Cybersecurity	The practice of protecting systems, networks, and data from cyberattacks, unauthorized access, and damage.	A bank uses firewalls, intrusion detection systems, and encryption to protect online banking transactions.
Cyberspace	The interconnected, virtual environment created by computers, networks, and digital communication systems.	Social media platforms, cloud storage, and online shopping sites are all part of cyberspace.
Confidentiality	Ensuring that only authorized people or systems can access sensitive information.	Encrypting patient medical records so only doctors can read them.
Integrity	Ensuring data remains accurate, complete, and unaltered unless changed by authorized processes.	Using checksums to detect changes in a downloaded file.
Availability	Ensuring information and resources are accessible when needed.	Using redundant data centers so a website remains online during a server failure.
Cyber Attack	A deliberate action by an adversary to compromise the confidentiality, integrity, or availability of a system or data.	A Distributed Denial of Service (DDoS) attack that makes an e-commerce site inaccessible.
Hacker	A skilled individual who explores computer systems; can be ethical or malicious.	A penetration tester identifying security flaws.
Cracker	Breaks into systems for malicious purposes.	Stealing credit card data from a retailer's database.
Noisy Hacker	Leaves obvious traces or disruption when hacking.	Defacing a company's homepage with graffiti.
Black Hat Hacker	Malicious hacker exploiting systems illegally.	Selling stolen login credentials on the dark web.
Gray Hat Hacker	Hacks without permission but without malicious intent; may report findings.	Discovering a website flaw and notifying the owner without prior authorization.
Ethical Hacker	Authorized professional testing systems to improve security.	Hired consultant performing a penetration test for a government agency.
Virus	Malicious code that attaches to files/programs and requires user action to spread.	The "ILOVEYOU" email virus spreading via infected attachments.
Worm	Self-replicating malware that spreads without user action, often over networks.	The "SQL Slammer" worm rapidly infected thousands of servers in 2003.
Trojan	Malicious software disguised as legitimate.	A fake "Flash Player update" installing spyware.
Rootkit	Software that hides malicious activities and grants administrator-level access.	A rootkit that hides a keylogger from antivirus scans.
Spyware	Software that secretly collects user information without consent.	A browser extension tracking all websites visited.
Ransomware	Malware that encrypts files and demands payment for decryption.	WannaCry attack encrypting hospital records in 2017.

Adware	Software that displays unwanted advertisements, sometimes tracking user activity.	A free game app constantly showing pop-up ads.
Crimeware	Malware designed for financial theft.	Zeus Trojan stealing online banking credentials.
Flaws/Bugs	Programming errors that may be exploited.	A buffer overflow in a web server allowing remote code execution.
Social Engineering	Manipulating people into revealing confidential information.	Pretending to be IT support to get a password.
DoS Attack	Overloading a system to make it unavailable.	Flooding a website with requests until it crashes.
DDoS Attack	Coordinated DoS attack using multiple compromised devices.	Botnet attack taking down Twitter in 2016.
Man-in-the-Middle (MITM)	Intercepting communication between two parties.	Intercepting online banking login details on an unsecured Wi-Fi network.
Phishing Attack	Fraudulent attempt to obtain sensitive info via deceptive messages.	Fake PayPal email asking for account verification.
Email Spoofing	Forging sender address in an email.	A spam email appearing to come from a trusted company.
IP Spoofing	Forging IP addresses to impersonate another device.	Sending packets with a forged source IP to bypass security filters.
Login Spoofing	Creating fake login interfaces to steal credentials.	Fake ATM screen asking for PIN.
Salami Attack	Stealing small amounts over time to avoid detection.	Bank employee transferring fractions of a cent from many accounts.
Email Bomb	Sending massive volumes of emails to crash a mailbox.	Flooding a company's HR inbox with thousands of junk emails.
Password Sniffing	Capturing passwords from network traffic.	Using packet sniffers to steal FTP credentials.
Buffer Overflow Attack	Overwriting memory to execute malicious code.	Exploiting a vulnerable C program to run injected shellcode.
Integer Overflow Attack	Causing a numeric variable to exceed its limit, triggering unexpected behavior.	Overflow in a shopping cart's quantity field causing free products.
Zero-Day Attack	Exploiting a vulnerability before the vendor releases a patch.	Exploit of an unpatched Windows SMB flaw.
Internal Theft	Theft of data/assets by an insider.	Employee copying customer database to a USB drive.
Session Hijacking	Taking over an active session between a user and a service.	Stealing a web session cookie to impersonate a user.
Web Jacking	Redirecting a legitimate website to a malicious one.	Altering DNS settings so visitors land on a phishing page.
SQL Injection	Injecting malicious SQL commands into input fields to manipulate a database.	Entering ' OR '1'='1 in a login form to bypass authentication.
Cross-Site Scripting (XSS)	Injecting malicious scripts into web pages viewed by others.	Inserting JavaScript into a forum post to steal session cookies.

Cross-Site Request Forgery (CSRF)	Forcing a logged-in user to perform unwanted actions.	Tricking a user into clicking a link that changes their email address in an account.
Brute Force Attack	Trying all possible passwords until the correct one is found.	Automated script trying millions of password combinations.
Rainbow Attack	Using precomputed hash tables to crack passwords.	Recovering a weak password hashed with MD5.
ARP Spoofing	Associating attacker's MAC address with victim's IP to intercept traffic.	Intercepting LAN traffic by poisoning ARP tables.
DNS Spoofing	Altering DNS records to redirect traffic.	Redirecting bank website visitors to a fake login page.
Wi-Fi Eavesdropping	Intercepting data transmitted over wireless networks.	Capturing passwords sent over open public Wi-Fi.